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Literature Review

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Article

Optical switching of a moiré Chern ferromagnet

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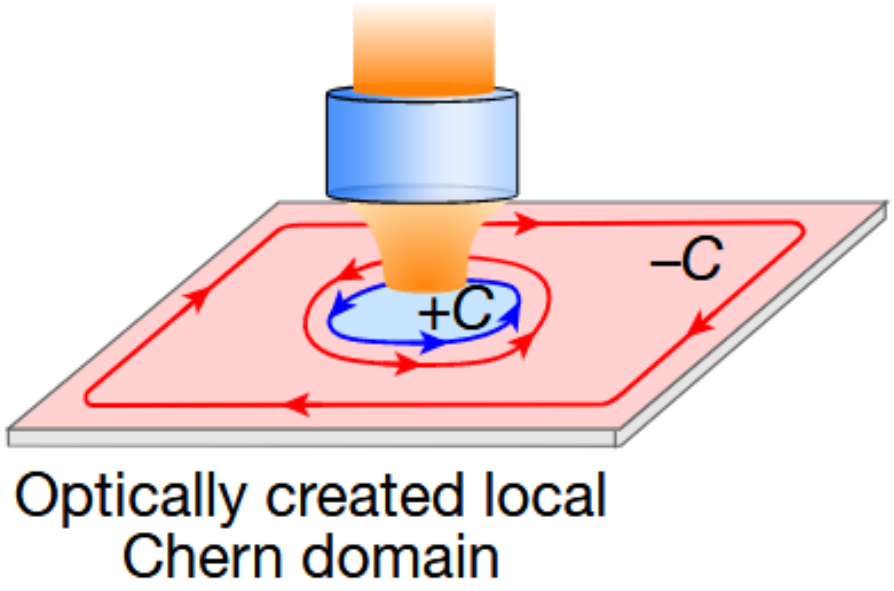
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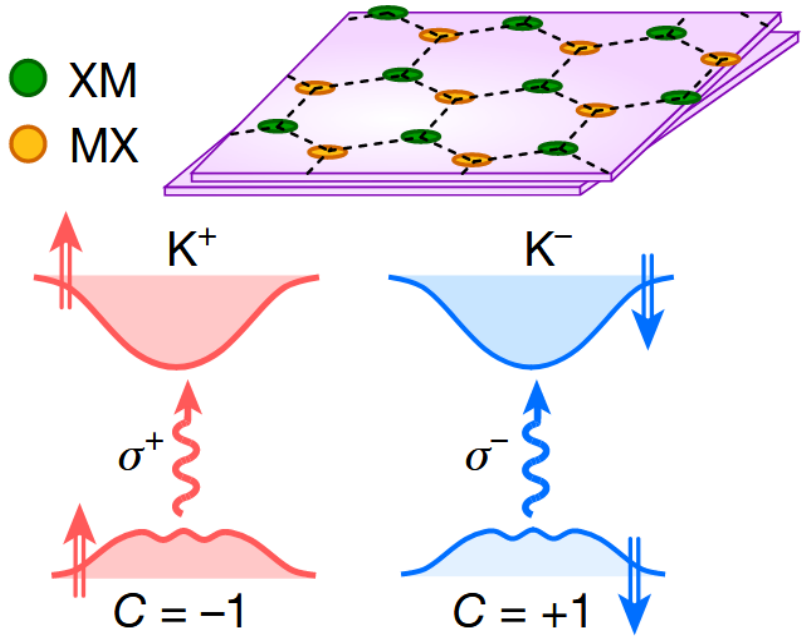
Check for updates

Xiangbin Cai[1,2,10], Haiyang Pan[1,2,10], Yuzhu Wang[2,10], Abdullah Rasmita[1,2], Shunshun Yang[1], Yan Zhao[1,2], Wei Wang[3], Ruihuan Duan[4], Ruihua He[1], Kenji Watanabe[5], Takashi Taniguchi[6], Zheng Liu[1,4], Jesús Zúñiga-Pérez[1,7], Bo Yang[2]✉ & Weibo Gao[1,2,8,9]✉

Optical Control over Chern Number

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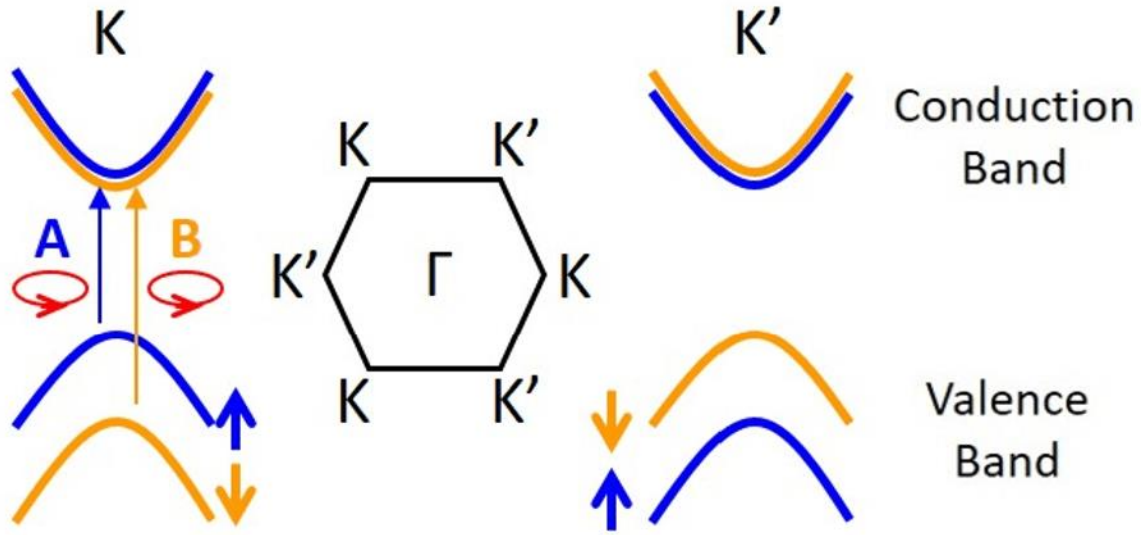


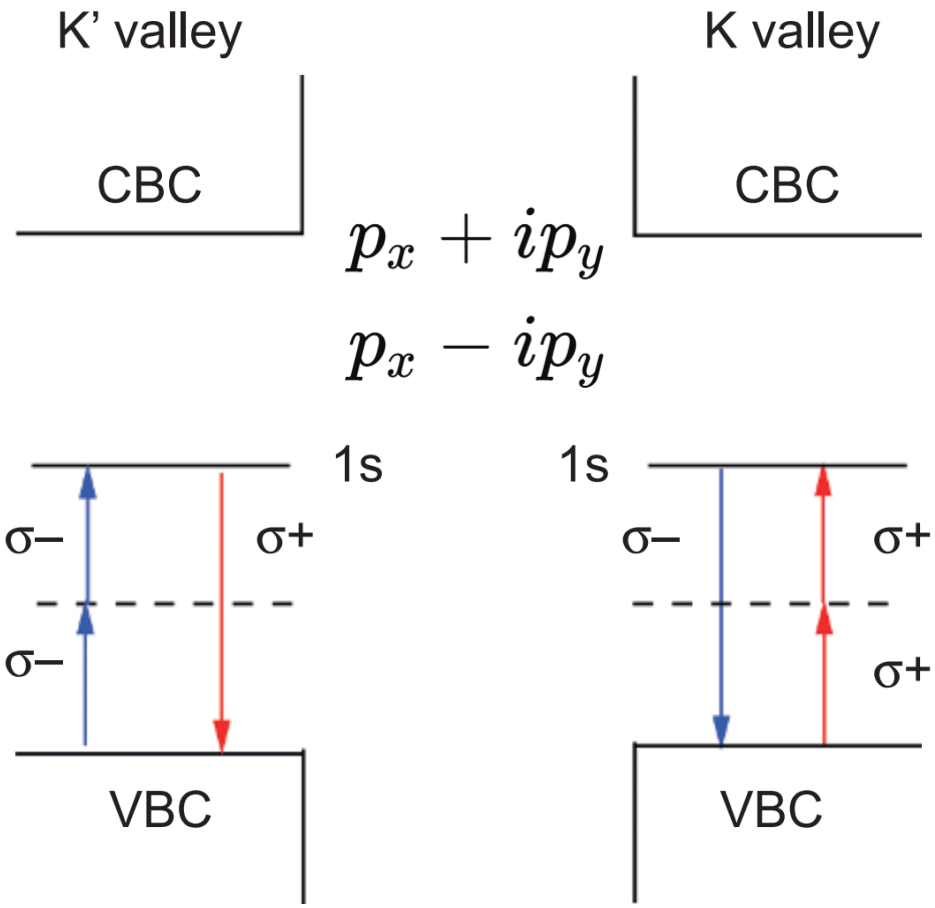


Nature 649, 1153–1158 (2026)

Spin-valley Locking and Polarization Selection

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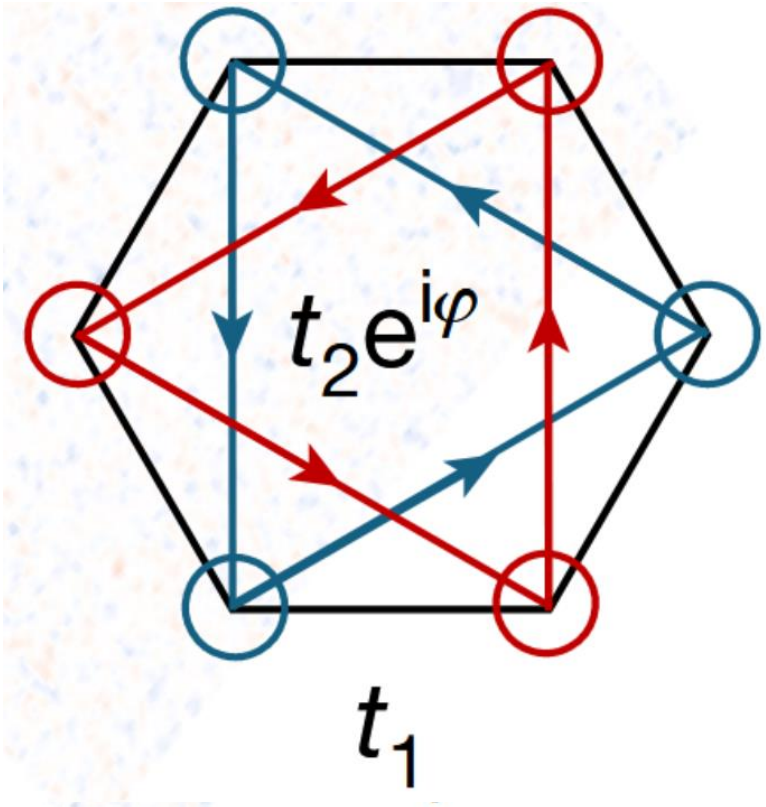


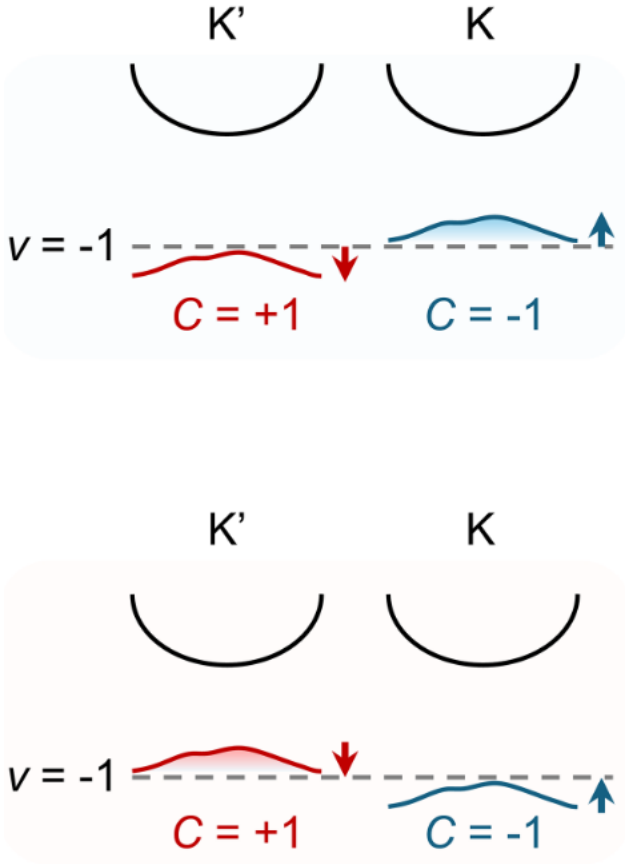


$$M_{\pm}(\mathbf{k}) \propto \langle c, \mathbf{k} | p_x \pm i p_y | v, \mathbf{k} \rangle$$

Light Sci Appl 4, e366 (2015)

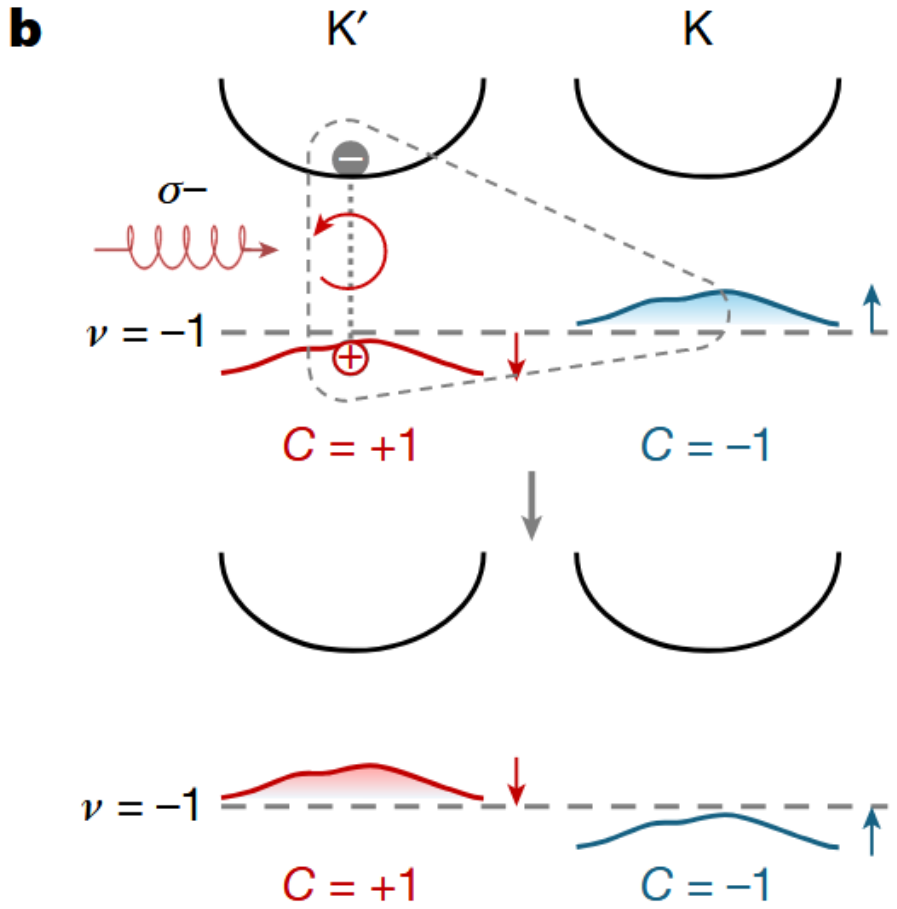
Moiré Chern Ferromagnet

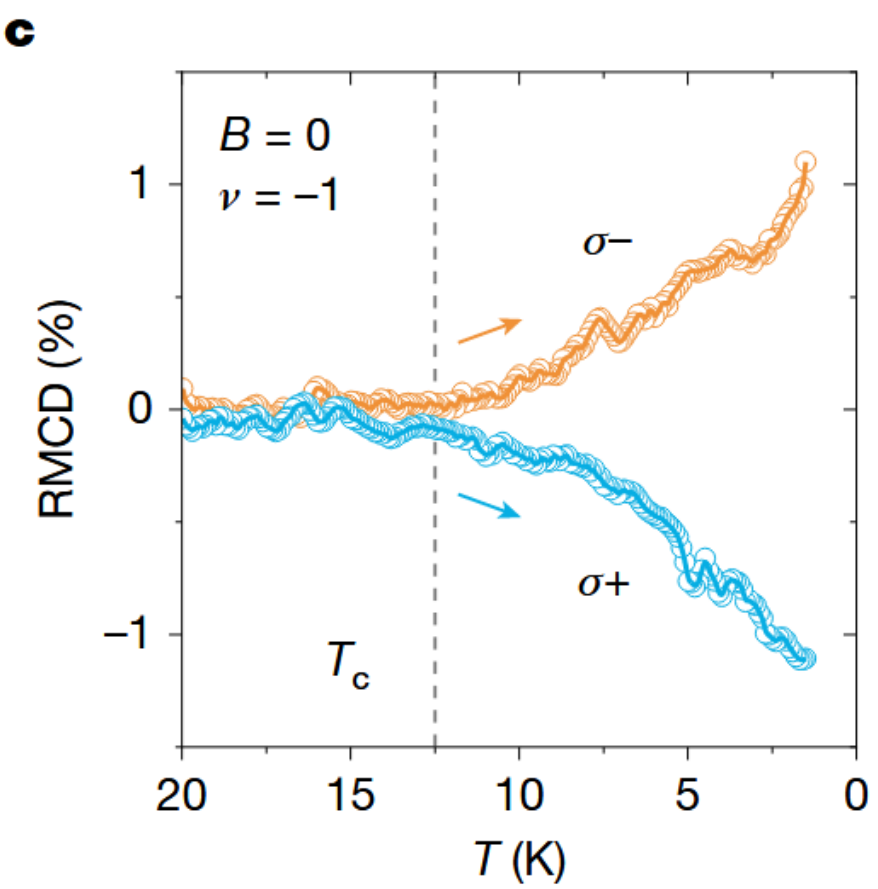




Nucleate Ferromagnetic Domain

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$$F(M) = a(T - T_c)M^2 + bM^4 - h_{\text{eff}}M$$

Landau Theory

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Near critical point, expand the free energy in powers of the spin-valley-coupled magnetization M

$$F(M;T) = F_0(T) + a(T)M^2 + bM^4 \ldots \quad (b > 0)$$

Hamiltonian is symmetric (no field) under spin direction flip (M→−M)

Temperature dependence

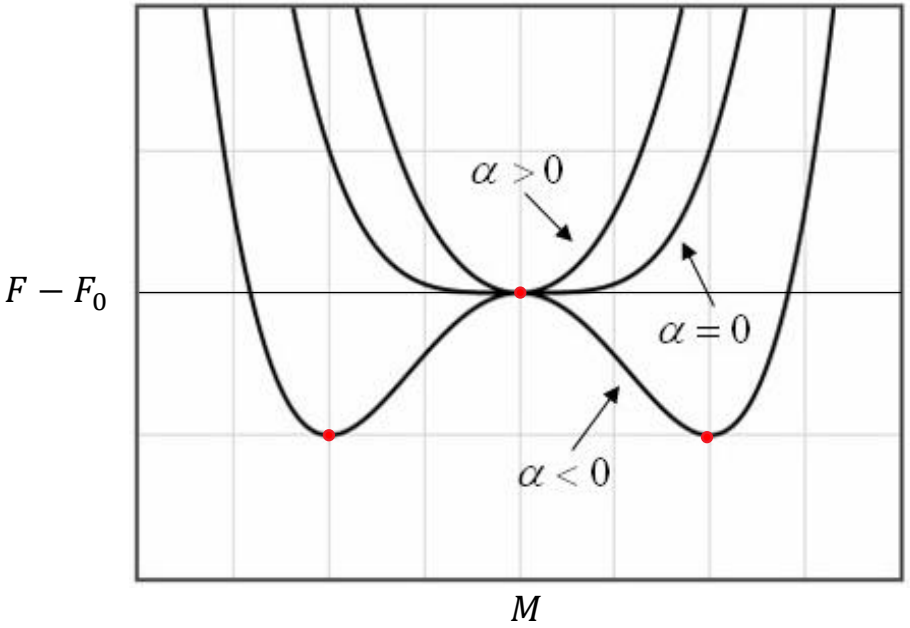
$T > T_C$: $a(T) > 0$, disorder (M=0)

$T = T_C$: $a(T) = 0$, second order transition point

$T < T_C$: $a(T) < 0$, order (M has non-zero solution)

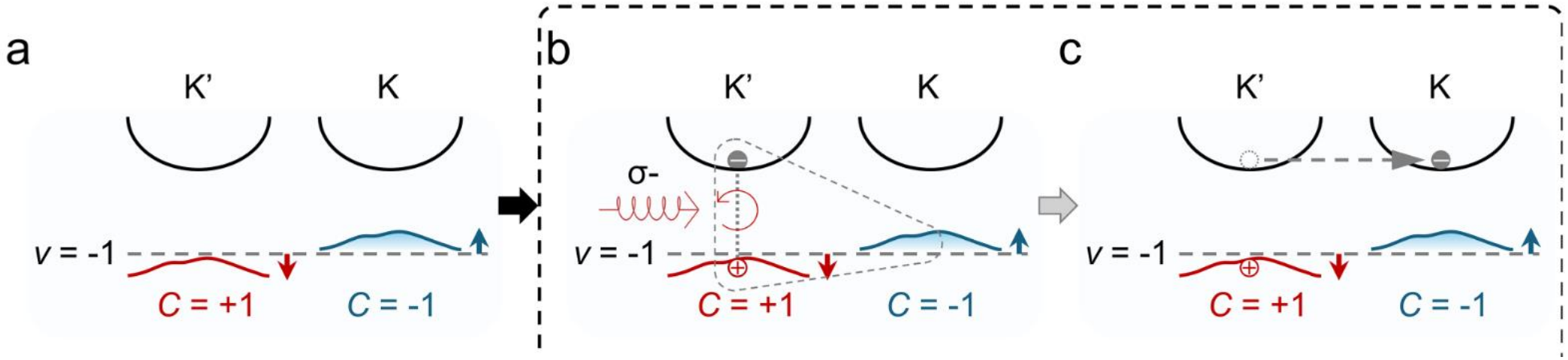
Define smooth $a(T) = a_0(T - T_C)$ $(a_0 > 0)$

$$F(M) = a(T - T_c)M^2 + bM^4 - h_{\text{eff}}M$$



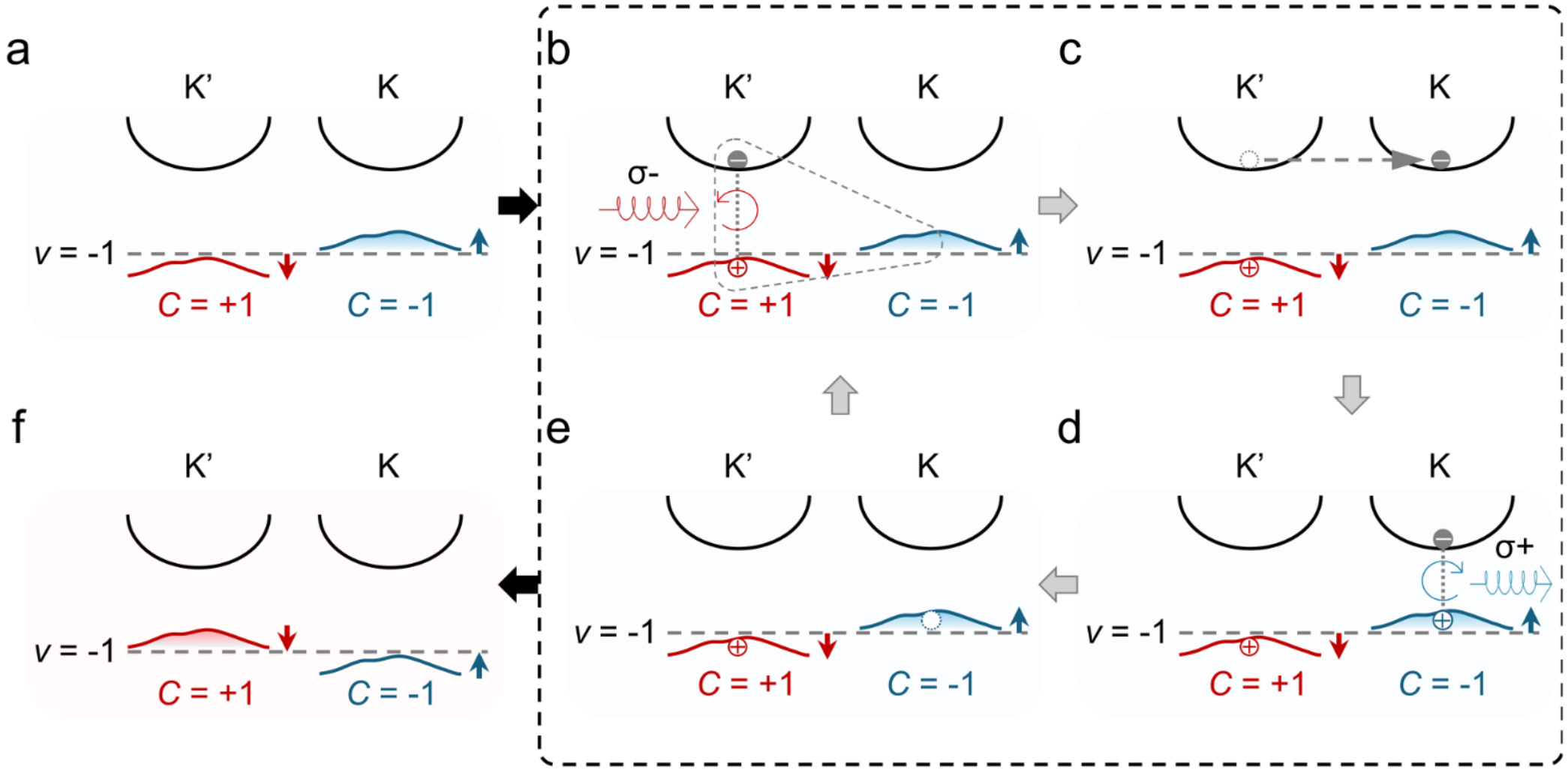
Mechanisms of Optical Switching

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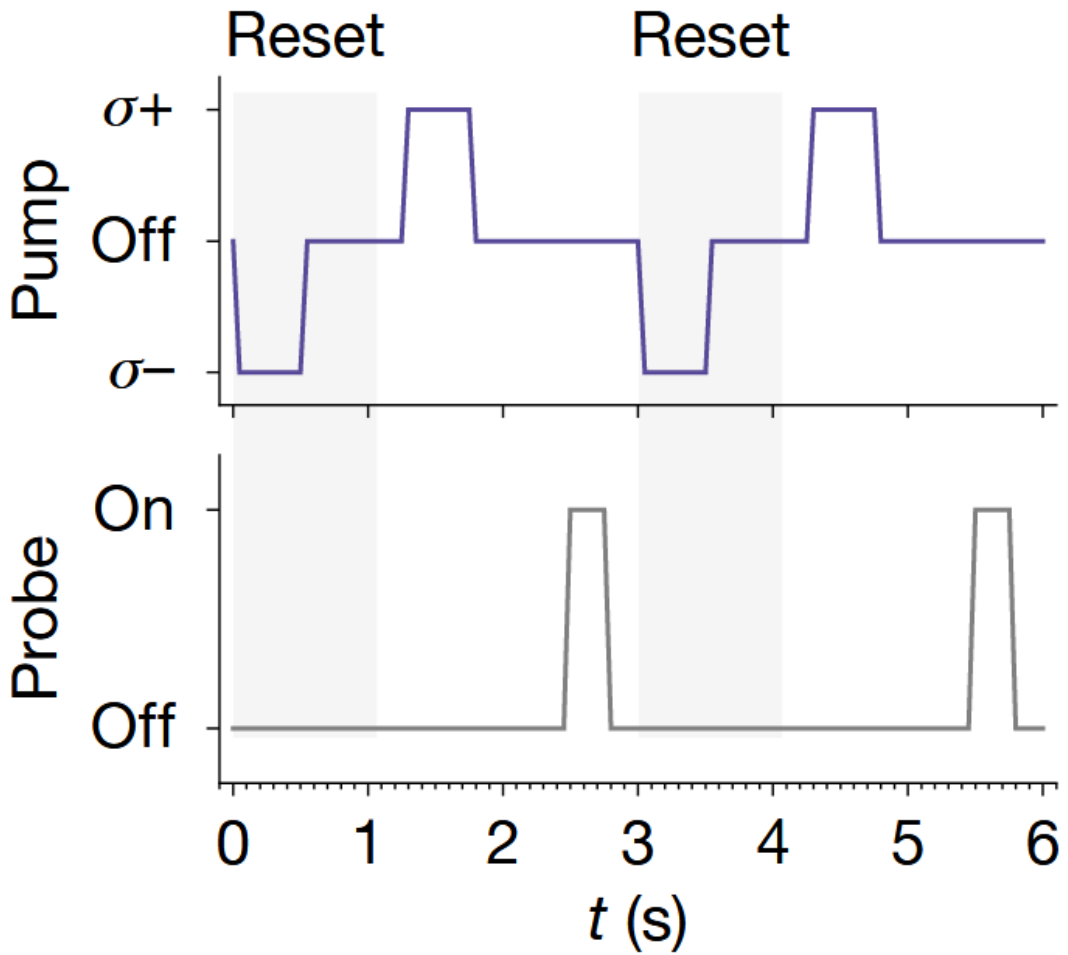
Mechanisms of Optical Switching

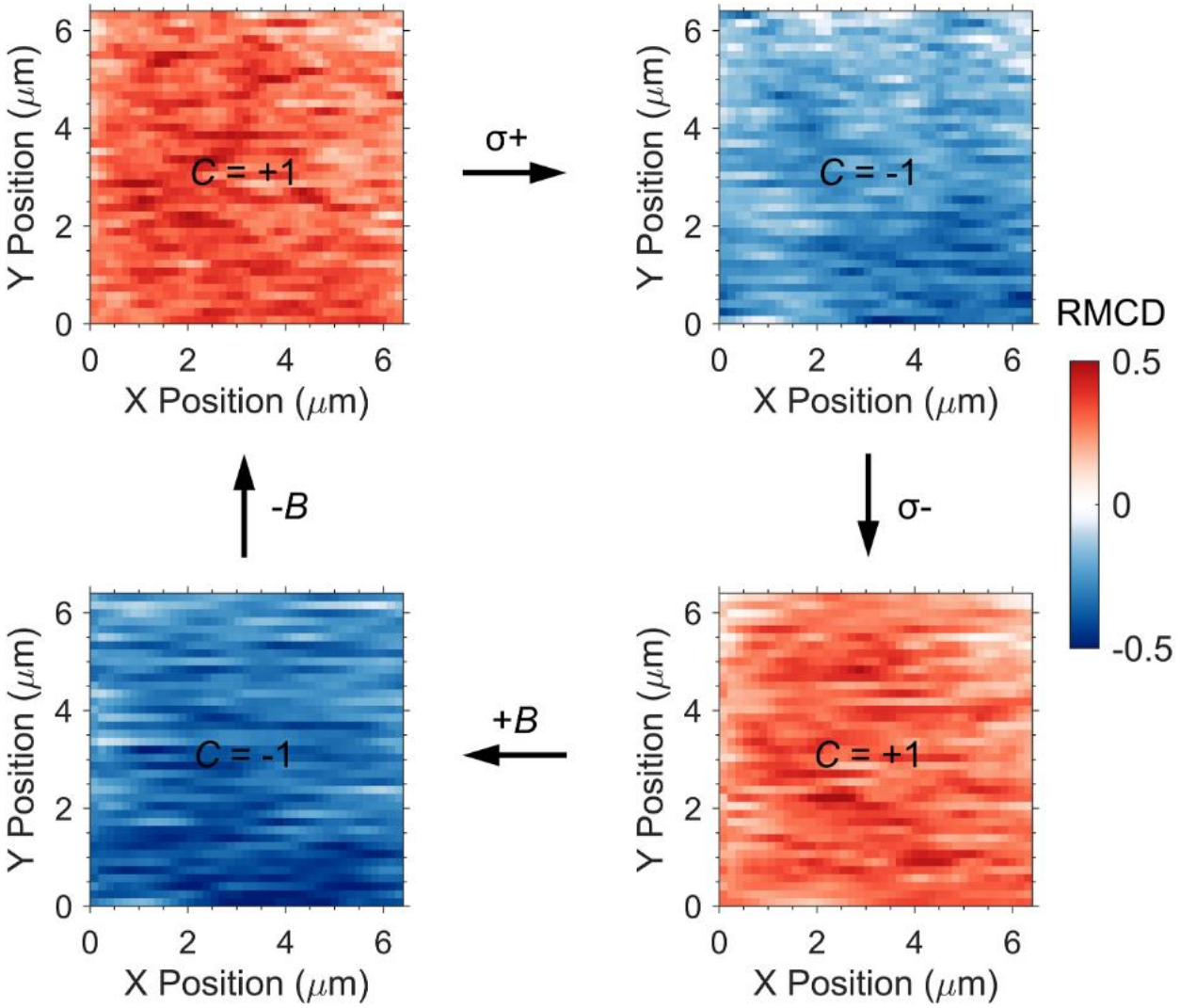
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Multiple Times during Continuous-Wave Pumping

Mechanisms of Optical Switching







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THANKS

Speaker: Zhongqi Xiu